

## SUCEAVA, Romania

WMO: 150230

Lat: 47.6323N

Lon: 26.2407E

Elev: 353

StdP: 97.15

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -15.8      | -13.0 | -18.0                         | 0.8 | -14.9 | -15.0 | 1.1 | -12.2 | 12.9                  | -8.8 | 11.5 | -5.5 | 2.8                   | 340  | 0.552 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 9.7                    | 30.2            | 20.2 | 28.6 | 19.9 | 27.0 | 19.3 | 22.2                | 26.9 | 21.1 | 26.5 | 20.1 | 25.2 | 3.7                  | 140  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 20.7                            | 16.1 | 25.0 | 19.2 | 14.6 | 23.4 | 18.2 | 13.7 | 22.3 | 66.7          | 26.8 | 63.0 | 26.3 | 59.7 | 25.0 | 26.2              |

## Extreme Annual Design Conditions

| Extreme Annual WS |     |      | Extreme Annual Temperature |       |      |                    | n-Year Return Period Values of Extreme Temperature |           |      |            |      |            |      |            |      |  |
|-------------------|-----|------|----------------------------|-------|------|--------------------|----------------------------------------------------|-----------|------|------------|------|------------|------|------------|------|--|
| 1%                |     | 2.5% | 5%                         | Mean  |      | Standard Deviation |                                                    | n=5 years |      | n=10 years |      | n=20 years |      | n=50 years |      |  |
| Min               |     | Max  |                            | Min   | Max  | Min                | Max                                                | Min       | Max  | Min        | Max  | Min        | Max  | Min        | Max  |  |
| (a)               | (b) | (c)  |                            | (e)   | (f)  | (g)                | (h)                                                | (i)       | (j)  | (k)        | (l)  | (m)        | (n)  | (o)        | (p)  |  |
| 10.2              | 9.0 | 8.0  | DB                         | -19.4 | 32.5 | 3.5                | 2.1                                                | -22.0     | 34.0 | -24.0      | 35.3 | -26.0      | 36.5 | -28.6      | 38.0 |  |
|                   |     |      | WB                         | -19.5 | 23.9 | 3.5                | 1.5                                                | -22.0     | 25.0 | -24.1      | 25.9 | -26.0      | 26.7 | -28.6      | 27.9 |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual      | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|-------------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)         | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 9.2         | -3.2  | -1.2  | 3.1   | 9.4   | 15.0  | 18.4  | 20.3  | 19.8  | 15.2  | 9.3   | 4.1   | -0.9  |  |  |
|                                                                        | DBStd       | 9.34        | 5.51  | 5.60  | 5.09  | 4.30  | 3.59  | 3.19  | 3.03  | 3.15  | 3.75  | 4.41  | 4.61  | 5.25  |  |  |
|                                                                        | HDD10.0     | 1601        | 410   | 313   | 218   | 61    | 6     | 0     | 0     | 0     | 5     | 66    | 184   | 338   |  |  |
|                                                                        | HDD18.3     | 3552        | 668   | 546   | 472   | 270   | 113   | 39    | 15    | 21    | 106   | 279   | 427   | 597   |  |  |
|                                                                        | CDD10.0     | 1295        | 0     | 0     | 4     | 42    | 160   | 252   | 320   | 304   | 161   | 46    | 7     | 0     |  |  |
|                                                                        | CDD18.3     | 206         | 0     | 0     | 0     | 1     | 9     | 41    | 77    | 67    | 11    | 1     | 0     | 0     |  |  |
|                                                                        | CDH23.3     | 1609        | 0     | 0     | 0     | 12    | 88    | 297   | 562   | 540   | 106   | 6     | 0     | 0     |  |  |
|                                                                        | CDH26.7     | 406         | 0     | 0     | 0     | 1     | 9     | 53    | 155   | 166   | 21    | 1     | 0     | 0     |  |  |
| (14) Wind                                                              | WSAvg       | 3.5         | 3.6   | 3.6   | 3.9   | 3.9   | 3.7   | 3.5   | 3.3   | 3.2   | 3.3   | 3.3   | 3.5   | 3.5   |  |  |
| (15) Precipitation                                                     | PrecAvg     | 632         | 20    | 23    | 31    | 51    | 75    | 97    | 105   | 77    | 54    | 45    | 29    | 26    |  |  |
|                                                                        | PrecMax     | 904         | 41    | 44    | 66    | 132   | 168   | 241   | 268   | 227   | 152   | 163   | 66    | 61    |  |  |
|                                                                        | PrecMin     | 402         | 6     | 7     | 9     | 13    | 22    | 32    | 14    | 29    | 11    | 8     | 2     | 2     |  |  |
|                                                                        | PrecStd     | 137         | 10    | 10    | 16    | 28    | 40    | 49    | 66    | 50    | 33    | 32    | 17    | 15    |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB          | 10.5  | 13.7  | 18.8  | 25.0  | 27.9  | 30.1  | 32.8  | 32.9  | 29.0  | 23.7  | 18.1  | 12.2  |  |  |
|                                                                        |             | MCWB        | 7.1   | 9.2   | 10.5  | 15.2  | 18.8  | 20.1  | 21.3  | 20.8  | 18.0  | 15.7  | 12.1  | 7.8   |  |  |
|                                                                        | 2%          | DB          | 7.3   | 10.1  | 15.2  | 21.3  | 25.8  | 28.1  | 30.2  | 30.2  | 26.0  | 20.8  | 15.0  | 9.1   |  |  |
|                                                                        |             | MCWB        | 5.0   | 6.7   | 9.4   | 13.3  | 17.0  | 20.1  | 20.6  | 20.0  | 17.4  | 14.6  | 10.8  | 6.3   |  |  |
|                                                                        | 5%          | DB          | 5.2   | 7.2   | 12.5  | 19.0  | 23.9  | 26.8  | 28.2  | 28.2  | 24.0  | 18.1  | 12.2  | 7.1   |  |  |
|                                                                        |             | MCWB        | 3.5   | 5.0   | 7.9   | 11.8  | 16.2  | 19.3  | 20.2  | 19.6  | 16.4  | 13.4  | 9.5   | 5.0   |  |  |
|                                                                        | 10%         | DB          | 3.8   | 5.4   | 10.1  | 16.8  | 21.9  | 25.0  | 26.7  | 26.5  | 21.9  | 16.0  | 10.5  | 5.2   |  |  |
|                                                                        |             | MCWB        | 2.4   | 3.8   | 6.6   | 10.9  | 15.2  | 18.5  | 19.8  | 19.1  | 15.6  | 12.4  | 8.6   | 3.7   |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4%        | WB          | 7.6   | 9.3   | 11.7  | 15.8  | 20.9  | 23.5  | 23.7  | 23.5  | 19.8  | 17.2  | 12.9  | 8.5   |  |  |
|                                                                        |             | MCDB        | 10.1  | 13.1  | 17.0  | 23.5  | 25.7  | 26.5  | 29.5  | 28.5  | 26.7  | 21.4  | 16.7  | 11.4  |  |  |
|                                                                        | 2%          | WB          | 5.2   | 6.9   | 9.8   | 14.1  | 18.2  | 21.5  | 22.1  | 22.0  | 18.4  | 15.3  | 11.5  | 6.7   |  |  |
|                                                                        |             | MCDB        | 6.8   | 9.6   | 14.2  | 20.2  | 23.0  | 25.9  | 27.1  | 26.7  | 23.7  | 19.0  | 14.2  | 8.7   |  |  |
|                                                                        | 5%          | WB          | 3.8   | 5.3   | 8.4   | 12.7  | 17.0  | 20.2  | 21.1  | 20.9  | 17.4  | 14.0  | 10.1  | 5.2   |  |  |
|                                                                        |             | MCDB        | 5.0   | 6.8   | 11.9  | 17.8  | 21.9  | 25.0  | 26.6  | 26.5  | 22.1  | 17.7  | 12.1  | 6.7   |  |  |
|                                                                        | 10%         | WB          | 2.3   | 3.9   | 7.0   | 11.4  | 15.9  | 19.2  | 20.1  | 19.7  | 16.4  | 12.7  | 8.8   | 3.9   |  |  |
|                                                                        |             | MCDB        | 3.4   | 5.2   | 9.7   | 15.8  | 20.6  | 24.0  | 25.2  | 24.6  | 20.6  | 15.5  | 10.5  | 5.2   |  |  |
| (35) Mean Daily Temperature Range                                      | 5% DB       | MDBR        | 5.6   | 6.2   | 7.8   | 9.5   | 9.9   | 9.5   | 9.7   | 10.0  | 9.3   | 8.6   | 6.5   | 5.6   |  |  |
|                                                                        |             | MCDBR       | 7.8   | 9.0   | 12.3  | 13.4  | 13.1  | 12.3  | 12.4  | 13.5  | 13.2  | 12.3  | 9.7   | 8.3   |  |  |
|                                                                        | 5% WB       | MCWBR       | 6.0   | 6.5   | 7.4   | 7.0   | 6.5   | 5.9   | 5.7   | 5.6   | 6.1   | 7.1   | 6.5   | 6.2   |  |  |
|                                                                        |             | MCWBR       | 7.2   | 8.4   | 11.2  | 12.2  | 11.4  | 10.6  | 10.8  | 11.7  | 11.3  | 10.9  | 8.9   | 7.7   |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |             | 0.310 | 0.341 | 0.379 | 0.439 | 0.414 | 0.412 | 0.439 | 0.446 | 0.401 | 0.365 | 0.335 | 0.308 |  |  |
|                                                                        |             | taud        | 2.311 | 2.252 | 2.225 | 2.093 | 2.241 | 2.279 | 2.210 | 2.184 | 2.268 | 2.352 | 2.378 | 2.354 |  |  |
|                                                                        | Edn at Noon |             | 755   | 806   | 831   | 811   | 851   | 853   | 822   | 796   | 799   | 772   | 721   | 712   |  |  |
|                                                                        |             | Edh at Noon | 74    | 98    | 118   | 149   | 133   | 129   | 137   | 135   | 113   | 89    | 70    | 64    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |             | 1.16  | 1.94  | 3.12  | 4.36  | 5.50  | 5.80  | 5.77  | 5.23  | 3.67  | 2.37  | 1.21  | 0.89  |  |  |
|                                                                        | RadStd      |             | 0.17  | 0.27  | 0.35  | 0.40  | 0.47  | 0.49  | 0.45  | 0.41  | 0.46  | 0.31  | 0.13  | 0.12  |  |  |

## Historical Trends

|      |                         | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------|-------------------------|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|      |                         |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) | Station Only            | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) | Regional (35 neighbors) | +0.69 | N/A     | N/A    | +0.90   | N/A   | N/A   | -148        | -211    | +103    | +32     |

Nomenclature: See separate page